1. Arrays & Strings (40 Questions)

Basic Traversal & Logic

- **LeetCode 1:** Two Sum (Easy)
- **LeetCode 121:** Best Time to Buy and Sell Stock (Easy)
- **LeetCode 26:** Remove Duplicates from Sorted Array (Easy)
- **LeetCode 27:** Remove Element (Easy)
- **LeetCode 169:** Majority Element (Easy)
- **HackerRank:** Arrays - Left Rotation (Easy)
- **LeetCode 58:** Length of Last Word (Easy)
- **LeetCode 88:** Merge Sorted Array (Easy)
- **LeetCode 28:** Find the Index of the First Occurrence in a String (Easy)
- **LeetCode 6:** Zigzag Conversion (Medium)
- **LeetCode 38:** Count and Say (Medium)

Two Pointers & Sliding Window

- **LeetCode 11:** Container With Most Water (Medium)
- **LeetCode 15:** 3Sum (Medium)
- **LeetCode 167:** Two Sum II - Input Array Is Sorted (Medium)
- **LeetCode 3:** Longest Substring Without Repeating Characters (Medium)
- **LeetCode 424:** Longest Repeating Character Replacement (Medium)
- **LeetCode 239:** Sliding Window Maximum (Hard)
- **LeetCode 125:** Valid Palindrome (Easy)
- **LeetCode 344:** Reverse String (Easy)
- **LeetCode 167:** Two Sum II - Input Array Is Sorted (Medium)
- **LeetCode 151:** Reverse Words in a String (Medium)
- **LeetCode 189:** Rotate Array (Medium)
- **LeetCode 443:** String Compression (Medium)
- **LeetCode 76:** Minimum Window Substring (Hard)

Prefix Sums & Hashing

- **LeetCode 238:** Product of Array Except Self (Medium)
- **LeetCode 560:** Subarray Sum Equals K (Medium)
- **LeetCode 49:** Group Anagrams (Medium)
- **LeetCode 128:** Longest Consecutive Sequence (Medium)
- **LeetCode 1:** Two Sum (Easy)
- **LeetCode 14:** Longest Common Prefix (Easy)

Matrix / 2D Arrays

- **LeetCode 73:** Set Matrix Zeroes (Medium)
- **LeetCode 54:** Spiral Matrix (Medium)
- **LeetCode 48:** Rotate Image (Medium)
- **LeetCode 43:** Multiply Strings (Medium)

Advanced String Manipulation

- **LeetCode 5:** Longest Palindromic Substring (Medium)
- **LeetCode 20:** Valid Parentheses (Easy)
- **HackerRank:** Sherlock and the Valid String (Medium)
- **HackerRank:** Special String Again (Medium)
- **LeetCode 76:** Minimum Window Substring (Hard)
- **LeetCode 42:** Trapping Rain Water (Hard)
- **LeetCode 31:** Next Permutation (Medium)
- **LeetCode 41:** First Missing Positive (Hard)

2. Hashing (20 Questions)

- **Frequency Tracking & Sets**
- **LeetCode 217:** Contains Duplicate (Easy)
- **LeetCode 242:** Valid Anagram (Easy)
- **LeetCode 383:** Ransom Note (Easy)
- **LeetCode 202:** Happy Number (Easy)

- **LeetCode 349:** Intersection of Two Arrays (Easy)
- **LeetCode 350:** Intersection of Two Arrays II (Easy)
- **HackerRank:** Hash Tables: Ransom Note (Easy)
- **HackerRank:** Two Strings (Easy)
- **Mapping Relationships**
- **LeetCode 205:** Isomorphic Strings (Easy)
- **LeetCode 290:** Word Pattern (Easy)
- **LeetCode 219:** Contains Duplicate II (Easy)
- **LeetCode 953:** Verifying an Alien Dictionary (Easy)
- **LeetCode 36:** Valid Sudoku (Medium)
- **LeetCode 454:** 4Sum II (Medium)
- **LeetCode 525:** Contiguous Array (Medium)
- **HackerRank:** Frequency Queries (Medium)
- **HackerRank:** Count Triplets (Medium)
- **Advanced Hashing Systems**
- **LeetCode 380:** Insert Delete GetRandom O(1) (Medium)
- **LeetCode 146:** LRU Cache (Medium)
- **LeetCode 460:** LFU Cache (Hard)

3. Searching & Sorting (20 Questions)

Standard Binary Search

- **LeetCode 704:** Binary Search (Easy)
- **LeetCode 278:** First Bad Version (Easy)
- **LeetCode 35:** Search Insert Position (Easy)
- **LeetCode 74:** Search a 2D Matrix (Medium)
- **LeetCode 34:** Find First and Last Position of Element in Sorted Array (Medium)
- **LeetCode 704:** Binary Search (Easy)

Modified Binary Search

- **LeetCode 33:** Search in Rotated Sorted Array (Medium)
- **LeetCode 81:** Search in Rotated Sorted Array II (Medium)
- **LeetCode 153:** Find Minimum in Rotated Sorted Array (Medium)
- **LeetCode 162:** Find Peak Element (Medium)
- **LeetCode 540:** Single Element in a Sorted Array (Medium)
- **LeetCode 240:** Search a 2D Matrix II (Medium)

Binary Search on Answer

- **LeetCode 875:** Koko Eating Bananas (Medium)
- **LeetCode 1011:** Capacity To Ship Packages Within D Days (Medium)
- **LeetCode 4:** Median of Two Sorted Arrays (Hard)

Sorting Mechanics & Quickselect

- **HackerRank:** Mark and Toys (Easy)
- **HackerRank:** Sorting: Bubble Sort (Easy)
- **HackerRank:** Sorting: Comparator (Medium)
- **LeetCode 215:** Kth Largest Element in an Array (Medium)
- **LeetCode 981:** Time Based Key-Value Store (Medium)
- **HackerRank:** Fraudulent Activity Notifications (Medium)

4. Linked Lists (20 Questions)

Basic Manipulations & Relinking

- **LeetCode 206:** Reverse Linked List (Easy)
- **LeetCode 83:** Remove Duplicates from Sorted List (Easy)
- **LeetCode 203:** Remove Linked List Elements (Easy)
- **LeetCode 234:** Palindrome Linked List (Easy)
- **HackerRank:** Insert a node at a specific position in a linked list (Easy)
- **LeetCode 92:** Reverse Linked List II (Medium)

- **LeetCode 61:** Rotate List (Medium)
- **LeetCode 86:** Partition List (Medium)

Multiple Pointers & Merging

- **LeetCode 21:** Merge Two Sorted Lists (Easy)
- **LeetCode 160:** Intersection of Two Linked Lists (Easy)
- **LeetCode 141:** Linked List Cycle (Easy)
- **LeetCode 142:** Linked List Cycle II (Medium)
- **LeetCode 19:** Remove Nth Node From End of List (Medium)
- **LeetCode 328:** Odd Even Linked List (Medium)
- **LeetCode 287:** Find the Duplicate Number (Medium)

Advanced Complex Lists

- **LeetCode 2:** Add Two Numbers (Medium)
- **LeetCode 138:** Copy List with Random Pointer (Medium)
- **LeetCode 143:** Reorder List (Medium)
- **LeetCode 148:** Sort List (Medium)
- **LeetCode 25:** Reverse Nodes in k-Group (Hard)

5. Stacks & Queues (20 Questions)

Basic Stack Operations

- **LeetCode 20:** Valid Parentheses (Easy)
- **LeetCode 232:** Implement Queue using Stacks (Easy)
- **LeetCode 225:** Implement Stack using Queues (Easy)
- **LeetCode 155:** Min Stack (Medium)
- **LeetCode 150:** Evaluate Reverse Polish Notation (Medium)
- **LeetCode 71:** Simplify Path (Medium)
- **HackerRank:** Balanced Brackets (Medium)
- **HackerRank:** Queues: A Tale of Two Stacks (Medium)

Monotonic Stacks

- **LeetCode 739:** Daily Temperatures (Medium)

- **LeetCode 853:** Car Fleet (Medium)
- **LeetCode 901:** Online Stock Span (Medium)
- **LeetCode 402:** Remove K Digits (Medium)
- **LeetCode 316:** Remove Duplicate Letters (Medium)
- **HackerRank:** Largest Rectangle (Medium)
- **LeetCode 84:** Largest Rectangle in Histogram (Hard)
- **HackerRank:** Min Max Riddle (Hard)

Parsing & Queues

- **LeetCode 394:** Decode String (Medium)
- **LeetCode 622:** Design Circular Queue (Medium)
- **LeetCode 224:** Basic Calculator (Hard)
- **LeetCode 239:** Sliding Window Maximum (Hard)

6. Trees & BSTs (25 Questions)

Tree Traversals & Depth Calculation

- **LeetCode 94:** Binary Tree Inorder Traversal (Easy)
- **LeetCode 104:** Maximum Depth of Binary Tree (Easy)
- **LeetCode 226:** Invert Binary Tree (Easy)
- **LeetCode 100:** Same Tree (Easy)
- **LeetCode 101:** Symmetric Tree (Easy)
- **HackerRank:** Tree: Height of a Binary Tree (Easy)

Structural Subtree Properties

- **LeetCode 543:** Diameter of Binary Tree (Easy)
- **LeetCode 110:** Balanced Binary Tree (Easy)
- **LeetCode 112:** Path Sum (Easy)
- **LeetCode 113:** Path Sum II (Medium)
- **LeetCode 1448:** Count Good Nodes in Binary Tree (Medium)

Breadth-First Level Logic

- **LeetCode 102:** Binary Tree Level Order Traversal (Medium)

- **LeetCode 103:** Binary Tree Zigzag Level Order Traversal (Medium)
- **LeetCode 199:** Binary Tree Right Side View (Medium)
- **LeetCode 116:** Populating Next Right Pointers in Each Node (Medium)

Binary Search Trees (BST)

- **LeetCode 108:** Convert Sorted Array to Binary Search Tree (Easy)
- **LeetCode 235:** Lowest Common Ancestor of a Binary Search Tree (Easy)
- **LeetCode 98:** Validate Binary Search Tree (Medium)
- **LeetCode 230:** Kth Smallest Element in a BST (Medium)
- **LeetCode 450:** Delete Node in a BST (Medium)

Tree Construction & Hard Ancestry

- **LeetCode 236:** Lowest Common Ancestor of a Binary Tree (Medium)
- **LeetCode 105:** Construct Binary Tree from Preorder and Inorder Traversal (Medium)
- **HackerRank:** Tree: Huffman Decoding (Medium)
- **LeetCode 124:** Binary Tree Maximum Path Sum (Hard)
- **LeetCode 297:** Serialize and Deserialize Binary Tree (Hard)
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7. Heaps (Priority Queues) (20 Questions)

Stream Processing & Top-K Problems

- **LeetCode 703:** Kth Largest Element in a Stream (Easy)
- **LeetCode 1046:** Last Stone Weight (Easy)
- **HackerRank:** QHEAP1 (Easy)
- **HackerRank:** Jesse and Cookies (Easy)
- **LeetCode 973:** K Closest Points to Origin (Medium)
- **LeetCode 215:** Kth Largest Element in an Array (Medium)
- **LeetCode 347:** Top K Frequent Elements (Medium)

Scheduling & Greedy Tracking

- **LeetCode 621:** Task Scheduler (Medium)
- **LeetCode 1094:** Car Pooling (Medium)

- **LeetCode 1834:** Single-Threaded CPU (Medium)
- **LeetCode 767:** Reorganize String (Medium)

K-Way Merges & Matrices

- **LeetCode 264:** Ugly Number II (Medium)
- **LeetCode 378:** Kth Smallest Element in a Sorted Matrix (Medium)
- **LeetCode 355:** Design Twitter (Medium)
- **LeetCode 23:** Merge k Sorted Lists (Hard)
- **LeetCode 632:** Smallest Range Covering Elements from K Lists (Hard)

Hard Heap Orchestration

- **LeetCode 295:** Find Median from Data Stream (Hard)
- **LeetCode 502:** IPO (Hard)
- **LeetCode 895:** Maximum Frequency Stack (Hard)
- **HackerRank:** Minimum Average Waiting Time (Hard)

8. Recursion & Backtracking (20 Questions)

Subsets, Permutations & Combinations

- **LeetCode 22:** Generate Parentheses (Medium)
- **LeetCode 77:** Combinations (Medium)
- **LeetCode 78:** Subsets (Medium)
- **LeetCode 90:** Subsets II (Medium)
- **LeetCode 46:** Permutations (Medium)
- **LeetCode 47:** Permutations II (Medium)
- **LeetCode 39:** Combination Sum (Medium)
- **LeetCode 40:** Combination Sum II (Medium)
- **HackerRank:** Recursive Digit Sum (Medium)

Grid Exploration & Parsing

- **LeetCode 17:** Letter Combinations of a Phone Number (Medium)
- **LeetCode 93:** Restore IP Addresses (Medium)

- **LeetCode 79:** Word Search (Medium)
- **LeetCode 131:** Palindrome Partitioning (Medium)
- **HackerRank:** Recursion: Davis' Staircase (Medium)
- **HackerRank:** Crossword Puzzle (Medium)

Advanced Constraint Satisfaction

- **LeetCode 51:** N-Queens (Hard)
- **LeetCode 52:** N-Queens II (Hard)
- **LeetCode 37:** Sudoku Solver (Hard)
- **LeetCode 10:** Regular Expression Matching (Hard)
- **LeetCode 44:** Wildcard Matching (Hard)
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9. Dynamic Programming (30 Questions)

1D DP & Basic Memoization

- **LeetCode 70:** Climbing Stairs (Easy)
- **LeetCode 746:** Min Cost Climbing Stairs (Easy)
- **LeetCode 198:** House Robber (Medium)
- **LeetCode 213:** House Robber II (Medium)
- **LeetCode 91:** Decode Ways (Medium)
- **HackerRank:** Max Array Sum (Medium)

Grid / 2D Matrix DP

- **LeetCode 62:** Unique Paths (Medium)
- **LeetCode 63:** Unique Paths II (Medium)
- **LeetCode 64:** Minimum Path Sum (Medium)

Knapsack & Change Target Strategies

- **LeetCode 322:** Coin Change (Medium)
- **LeetCode 518:** Coin Change II (Medium)
- **LeetCode 416:** Partition Equal Subset Sum (Medium)
- **LeetCode 494:** Target Sum (Medium)

Subarrays & Longest Sequences

- **LeetCode 152:** Maximum Product Subarray (Medium)
- **LeetCode 300:** Longest Increasing Subsequence (Medium)
- **LeetCode 1143:** Longest Common Subsequence (Medium)
- **LeetCode 139:** Word Break (Medium)
- **HackerRank:** Abbreviation (Medium)

State Optimization & Stock Variants

- **LeetCode 122:** Best Time to Buy and Sell Stock II (Medium)
- **LeetCode 309:** Best Time to Buy and Sell Stock with Cooldown (Medium)
- **LeetCode 714:** Best Time to Buy and Sell Stock with Transaction Fee (Medium)
- **LeetCode 123:** Best Time to Buy and Sell Stock III (Hard)
- **LeetCode 188:** Best Time to Buy and Sell Stock IV (Hard)

String Intervals & Hard DP

- **LeetCode 5:** Longest Palindromic Substring (Medium)
- **LeetCode 647:** Palindromic Substrings (Medium)
- **LeetCode 32:** Longest Valid Parentheses (Hard)
- **LeetCode 72:** Edit Distance (Hard)
- **LeetCode 87:** Scramble String (Hard)
- **LeetCode 115:** Distinct Subsequences (Hard)
- **LeetCode 312:** Burst Balloons (Hard)

10. Graphs (25 Questions)

Matrix DFS & BFS Traversals

- **LeetCode 200:** Number of Islands (Medium)
- **LeetCode 547:** Number of Provinces (Medium)
- **LeetCode 695:** Max Area of Island (Medium)
- **LeetCode 130:** Surrounded Regions (Medium)
- **LeetCode 417:** Pacific Atlantic Water Flow (Medium)

- **LeetCode 994:** Rotting Oranges (Medium)
- **LeetCode 841:** Keys and Rooms (Medium)

Cycles & Disjoint Sets (Union-Find)

- **LeetCode 133:** Clone Graph (Medium)
- **LeetCode 684:** Redundant Connection (Medium)
- **LeetCode 785:** Is Graph Bipartite? (Medium)
- **LeetCode 990:** Satisfiability of Equality Equations (Medium)

Topological Sort & Dependencies

- **LeetCode 207:** Course Schedule (Medium)
- **LeetCode 210:** Course Schedule II (Medium)
- **LeetCode 802:** Find Eventual Safe States (Medium)

Shortest Paths & MST

- **LeetCode 399:** Evaluate Division (Medium)
- **LeetCode 743:** Network Delay Time (Medium)
- **LeetCode 1584:** Min Cost to Connect All Points (Medium)
- **HackerRank:** BFS: Shortest Reach in a Graph (Medium)
- **HackerRank:** Find the nearest clone (Medium)

Hard Multi-state Graph Problems

- **LeetCode 126:** Word Ladder II (Hard)
- **LeetCode 127:** Word Ladder (Hard)
- **LeetCode 332:** Reconstruct Itinerary (Hard)
- **LeetCode 778:** Swim in Rising Water (Hard)
- **LeetCode 815:** Bus Routes (Hard)
- **HackerRank:** Matrix (Hard)

11. Greedy Algorithms (20 Questions)

Basic Linear Choices

- **LeetCode 605:** Can Place Flowers (Easy)

- **HackerRank:** Max Min (Easy)
- **LeetCode 53:** Maximum Subarray (Medium)
- **LeetCode 134:** Gas Station (Medium)
- **LeetCode 846:** Hand of Straights (Medium)
- **LeetCode 1578:** Minimum Time to Make Rope Colorful (Medium)
- **LeetCode 1899:** Merge Triplets to Form Target Triplet (Medium)

Intervals & Boundary Optimization

- **LeetCode 55:** Jump Game (Medium)
- **LeetCode 45:** Jump Game II (Medium)
- **LeetCode 56:** Merge Intervals (Medium)
- **LeetCode 57:** Insert Interval (Medium)
- **LeetCode 435:** Non-overlapping Intervals (Medium)
- **LeetCode 452:** Minimum Number of Arrows to Burst Balloons (Medium)
- **LeetCode 763:** Partition Labels (Medium)
- **HackerRank:** Greedy Florist (Medium)

Queue & String Sequencing

- **LeetCode 406:** Queue Reconstruction by Height (Medium)
- **LeetCode 678:** Valid Parenthesis String (Medium)
- **LeetCode 135:** Candy (Hard)
- **LeetCode 330:** Patching Array (Hard)
- **HackerRank:** Reverse Shuffle Merge (Hard)

12. Bit Manipulation & Math (20 Questions)

Bitwise Logic & Arithmetic Tricking

- **LeetCode 136:** Single Number (Easy)
- **LeetCode 190:** Reverse Bits (Easy)
- **LeetCode 191:** Number of 1 Bits (Easy)
- **LeetCode 268:** Missing Number (Easy)

- **LeetCode 338:** Counting Bits (Easy)
- **HackerRank:** Lonely Integer (Easy)
- **HackerRank:** Flipping bits (Easy)
- **HackerRank:** Sum vs XOR (Easy)
- **LeetCode 371:** Sum of Two Integers (Medium)

Masking & Bit Arrays

- **LeetCode 137:** Single Number II (Medium)
- **LeetCode 260:** Single Number III (Medium)
- **LeetCode 201:** Bitwise AND of Numbers Range (Medium)
- **LeetCode 89:** Gray Code (Medium)
- **LeetCode 1980:** Find Unique Binary String (Medium)

Math Modulo & Primes

- **LeetCode 476:** Number Complement (Easy)
- **LeetCode 7:** Reverse Integer (Medium)
- **LeetCode 29:** Divide Two Integers (Medium)
- **LeetCode 48:** Rotate Image (Medium)
- **LeetCode 50:** Pow(x, n) (Medium)
- **LeetCode 204:** Count Primes (Medium)

13. Tries (20 Questions)

Foundational Trie Structure

- **LeetCode 14:** Longest Common Prefix (Easy)
- **HackerRank:** No Prefix Set (Easy)
- **LeetCode 208:** Implement Trie (Prefix Tree) (Medium)
- **LeetCode 211:** Design Add and Search Words Data Structure (Medium)
- **LeetCode 648:** Replace Words (Medium)
- **HackerRank:** Contacts (Medium)

Prefix String Lookups

- **LeetCode 676:** Implement Magic Dictionary (Medium)
- **LeetCode 677:** Map Sum Pairs (Medium)
- **LeetCode 720:** Longest Word in Dictionary (Medium)
- **LeetCode 792:** Number of Matching Subsequences (Medium)
- **LeetCode 1233:** Remove Subfolders from the Filesystem (Medium)
- **LeetCode 1268:** Search Suggestions System (Medium)

Bitwise Tries & Hard String Arrays

- **LeetCode 421:** Maximum XOR of Two Numbers in an Array (Medium)
- **LeetCode 212:** Word Search II (Hard)
- **LeetCode 336:** Palindrome Pairs (Hard)
- **LeetCode 472:** Concatenated Words (Hard)
- **LeetCode 745:** Prefix and Suffix Search (Hard)
- **LeetCode 1032:** Stream of Characters (Hard)
- **LeetCode 1707:** Maximum XOR With an Element From Array (Hard)
- **LeetCode 2416:** Sum of Prefix Scores of Strings (Hard)